

1 INGEST AND REPRESENT



Learner
uploads a lecture



Lecture video / PDF
DOCX and PPTX too



Dual channel extraction
EasyOCR + Faster Whisper



Fused timeline
board text + speech, timed



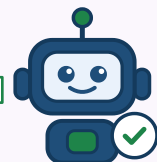
Summary and index
MiniLM embeddings + FAISS

personalised quiz returns to the learner

2 PLAN, GENERATE AND VERIFY



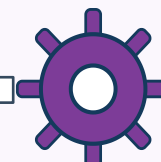
Formatter and Grader
gemma3:12b



Validator
gemma3:4b



Generator
gemma3:4b + search_lecture



Controller
recall, inference, synthesis



Quiz Planner
gemma3:4b

Refiner: regenerate, fetch context or replan

search_lecture retrieval tool

Ollama runtime, fully local

gemma3:4b generator, pinned

gemma3:12b judge and grader

MiniLM + FAISS retrieval

single NVIDIA T4, no paid API